

Tests on running inference on the mediapipe .tflite obd model inference tool web for l3f

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Situation overview

This document is a summary of tests run on the mediapipe .tflite obd model inference web tool I built for the l3f project. It aims to confirm the web tool is reliable & usable across multiple hardware options without crashing & misbehavior. This is a summary of multiple tests

Setup

Tests For every test we record the following environment variables:

- device
- OS
- browser + version
- camera
- target sampling rate
- model
- delegate
- score threshold
- goal of test
- duration of test
- special notes on env

, where

- **browser + version** is (index.html -> Environment -> Browser) rendering
- **target sampling rate** is the fps parameter given to `startCameraInferenceLoop()` when called from within `handleStartCamera()` in `src/main.js`
- **model** is: DIRNAME where `data_and_training/models/DIRNAME/model.tflite` is identical to `browser_app/public/models/model.tflite`; Test: `cmp browser_app/public/models/model.tflite data_and_training/models/DIRNAME/model.tflite -> silent exit, no browser_app/public/models/model.tflite data_and_training/models/DIRNAME/model.tflite differ: byte xx, line y type output.`
- **delegate** is the `delegate` parameter given to `ObjectDetector.createFromOptions` in model loading/compilation section of `try` block of `startApplication()` function in `src/main.js`
- **score threshold** is the `scoreThreshold` parameter given to `ObjectDetector.createFromOptions` in model loading/compilation section of `try` block of `startApplication()` function in `src/main.js`

and we collect

- csv formatted measurements (camera fpc, sampling fps, inference fps, inference latency) & appropriate visualisations
- UI behavior observations
- browser command line prints of note
- other notes

, where

browser command line prints of note means we ignore per-detection/per-measurement prints, but note any execution time errors.

Technical Render this document by running `Rscript -e "rmarkdown::render('reporting.Rmd')"` From `browser_app/measurements/`

Libraries:

```
# For data manipulation
suppressPackageStartupMessages(library(dplyr)) # <- Don't print warning on import
library(tidyr)

# For plotting & rendering
library(ggplot2)
library(scales)
```

Diagram drawing function (edited for Rmd use from `data_and_training/devtools/fpsdiagram.R`):

```
fpsdiagram <- function(input_filename) {

  if (!file.exists(input_filename)) {
    stop(
      paste0("Error: File '", input_filename, "' not found."),
      call. = FALSE
    )
  }

  df <- read.csv(input_filename, stringsAsFactors = FALSE)

  df$Timestamp <- as.POSIXct(
    df$Timestamp,
    format = "%Y-%m-%dT%H:%M:%OSZ",
    tz = "UTC"
  )

  df_long <- df %>%
    pivot_longer(
      cols = c(Camera_FPS, Sampling_FPS, Inference_FPS),
      names_to = "Metric",
      values_to = "FPS"
    )

  p <- ggplot(
    df_long,
    aes(x = Timestamp, y = FPS, color = Metric)
  ) +
    geom_line(linewidth = 1) +
    geom_point(size = 2) +
    scale_x_datetime(labels = date_format("%H:%M:%S")) +
    labs(
      title = "Performance Monitoring Metrics",
      subtitle = paste("File:", input_filename),
      x = "Timestamp (UTC)",
      y = "FPS",
      color = "Metric"
    ) +
```

```

theme_minimal(base_size = 12) +
theme(
  legend.position = "bottom",
  axis.text.x = element_text(angle = 45, hjust = 1)
)

print(p)
}

```

Test series 1: Finding the optimal sampling rate

Test 1.1 starting from a low baseline

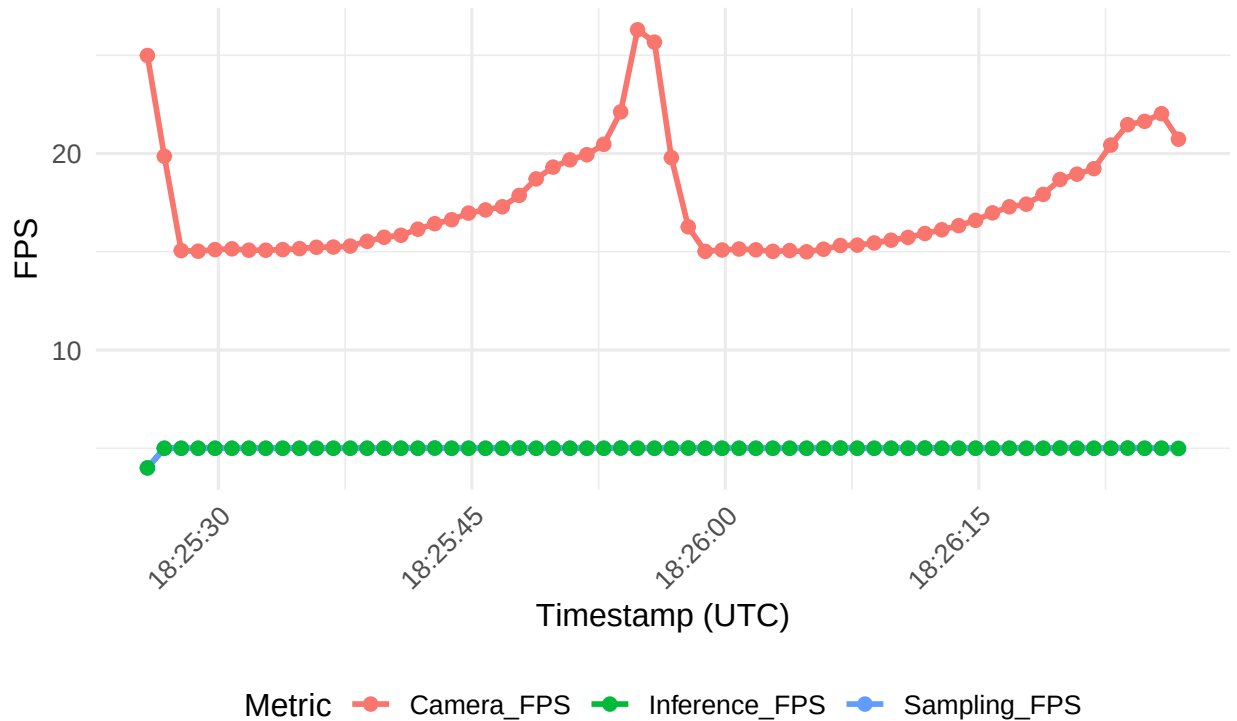
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 5
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establishment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr5_20260921_212518.csv")
```

Performance Monitoring Metrics

File: fps_sr5_20260921_212518.csv



Results

No record of browser command line prints.

Notes: Inference_FPS matches Sampling_FPS, significantly below Camera_FPS, can raise.

Test 1.2 raising to 10 fps sampling

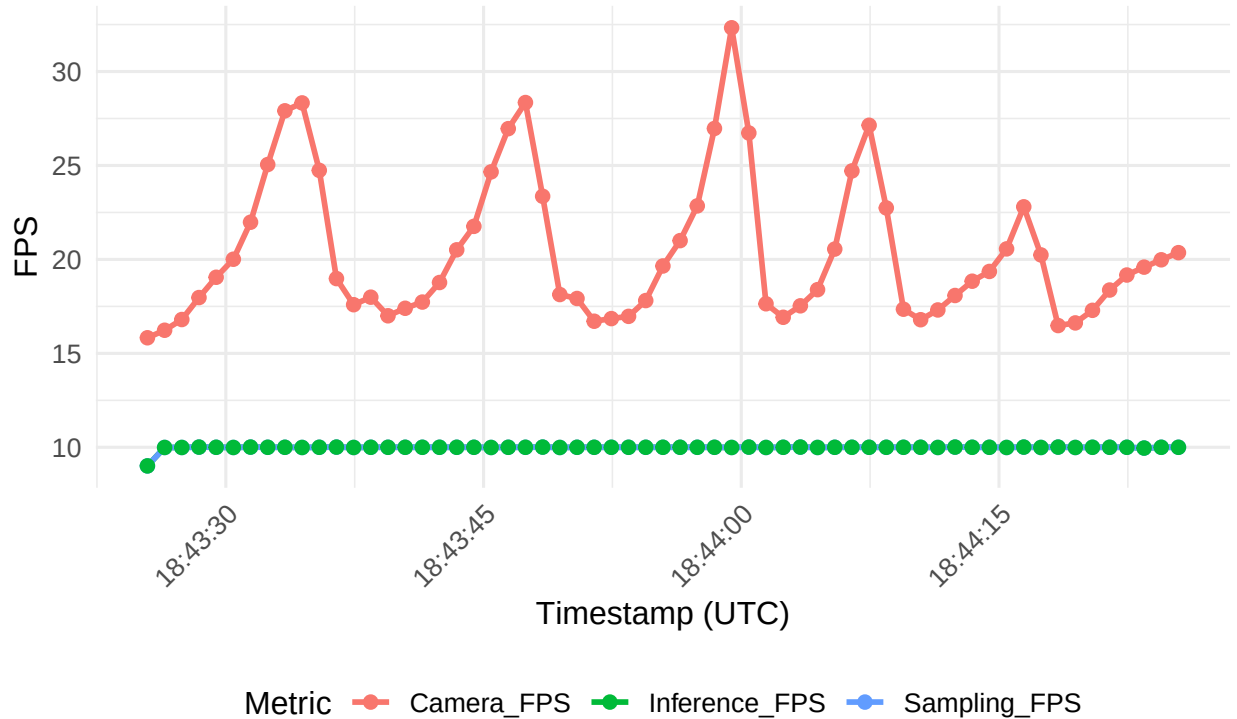
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 10
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establishment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr10_20260921_214321.csv")
```

Performance Monitoring Metrics

File: fps_sr10_20260921_214321.csv



Results

No record of browser command line prints.

Notes: Inference_FPS matches Sampling_FPS, still below Camera_FPS, can raise.

Test 1.3 reaching camera fps

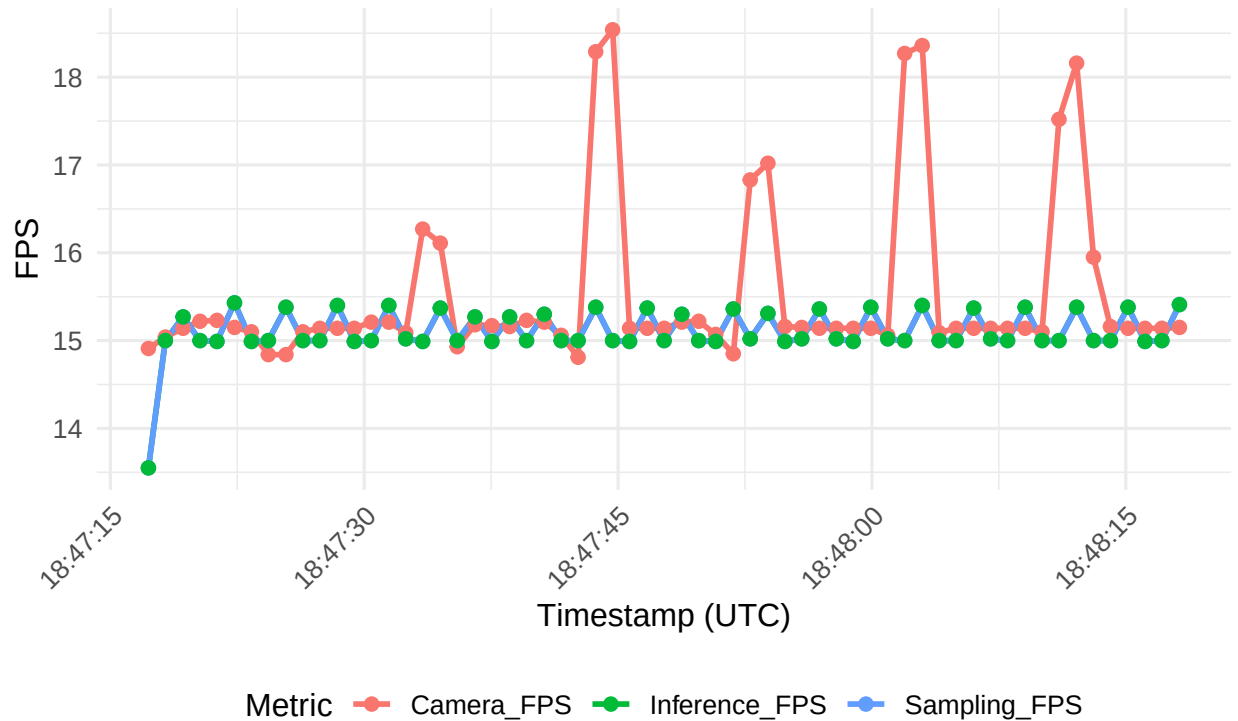
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 15
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establishment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr15_20260921_214710.csv")
```

Performance Monitoring Metrics

File: fps_sr15_20260921_214710.csv



Results

No record of browser command line prints.

Notes: Inference_FPS & Sampling_FPS now occasionally above Camera_FPS, will lower.

Test 1.4 achieving stability

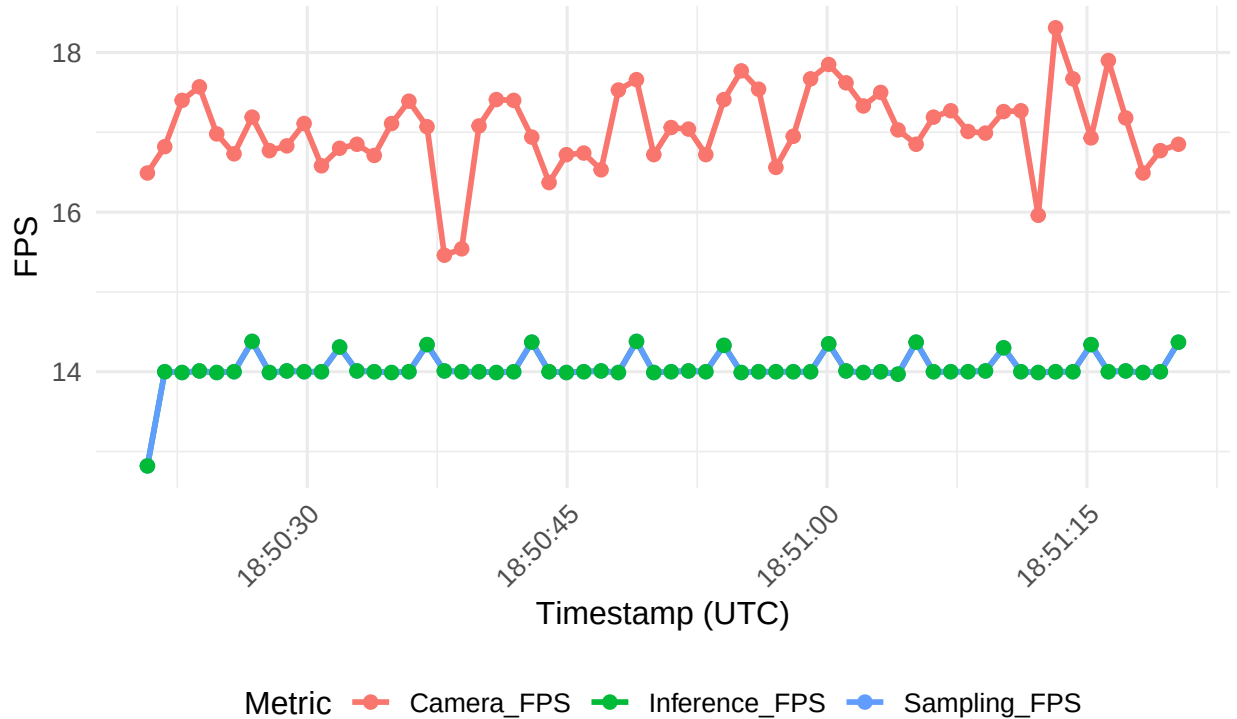
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establishment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr14_20260921_215016.csv")
```

Performance Monitoring Metrics

File: fps_sr14_20260921_215016.csv



Results

No record of browser command line prints.

Notes: Inference_FPS & Sampling_FPS now stable below Camera_FPS, not surpassing it. NOTE: Latency_MS (how long it actually takes to complete an inference) ranges from 46 to 49. It is still realistic to raise it. We're bottlenecked by Camera_FPS on this hardware.

Test series 2: On different devices

Test 2.1 on an old laptop (Thinkpad T420)

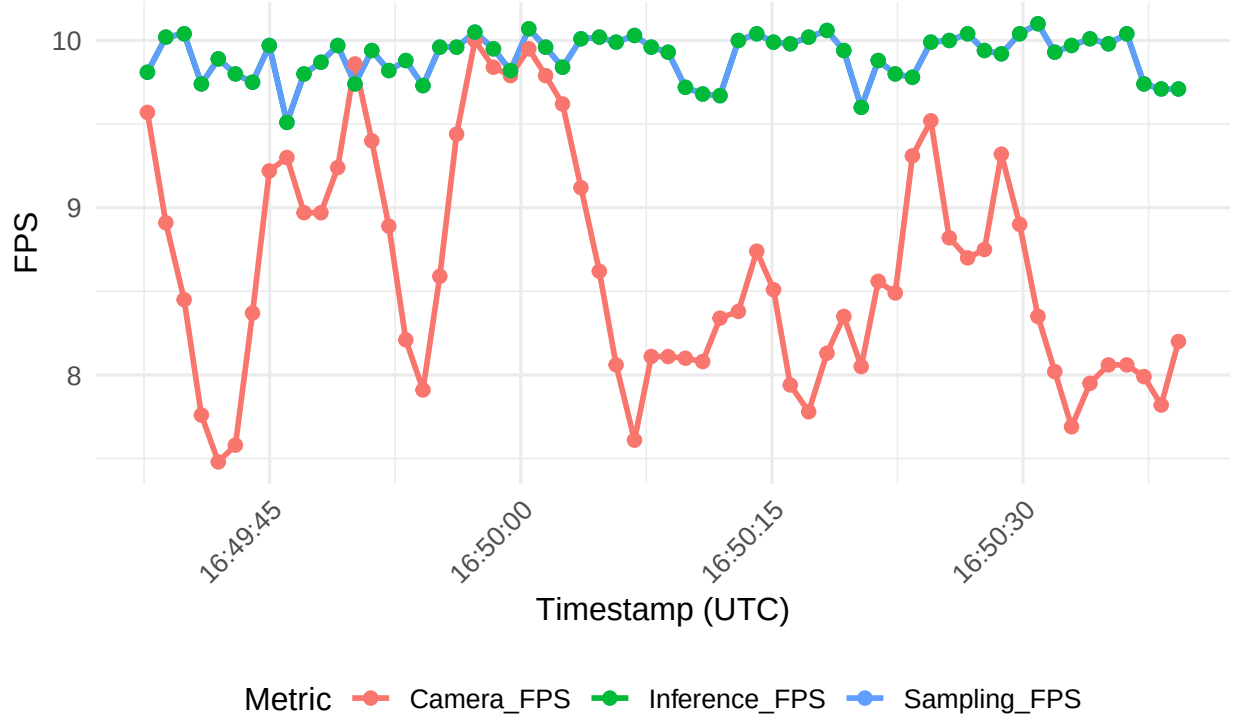
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Integrated Camera
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_l3f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: baseline on laptop
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_20260922_194937.csv")
```

Performance Monitoring Metrics

File: fps_t420_20260922_194937.csv



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

Notes: Camera_FPS ranges from below 8 to just around 10. Sampling_FPS & Inference_FPS hovers around 10. Latency_MS ranges from 97 to 103. The limiting factor with this setup is still Camera_FPS, but it's noticeably lower than on other computer. Will set sampling rate to 7 to achieve stability.

Test 2.2 T420, lowered sample rate

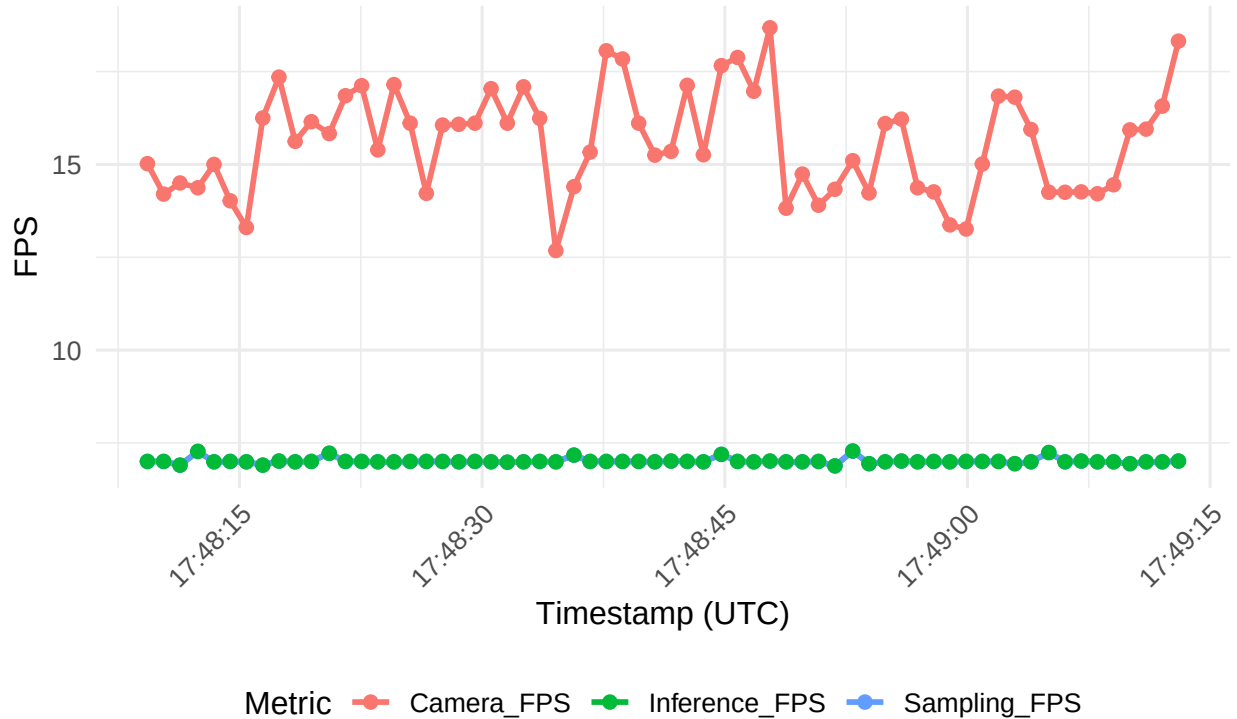
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Integrated Camera
- target sampling rate: 7
- model: ssd_plus_mobilenet_v2_13f_20260807_225848
- delegate: CPU
- score threshold: 0.35
- goal of test: baseline on laptop
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_sr7_20260922_204808.csv")
```

Performance Monitoring Metrics

File: fps_t420_sr7_20260922_204808.csv



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

Notes: **Camera_FPS** suddenly jumped to around 15. So the limiting factor probably wasn't the **Camera_FPS** but the inference. Inferences now range from 98 to 114, which is unusual. Trying to reach a slightly better baseline at 8 fps.

Test 2.3 T420, attempting to reach a better baseline at 8fps

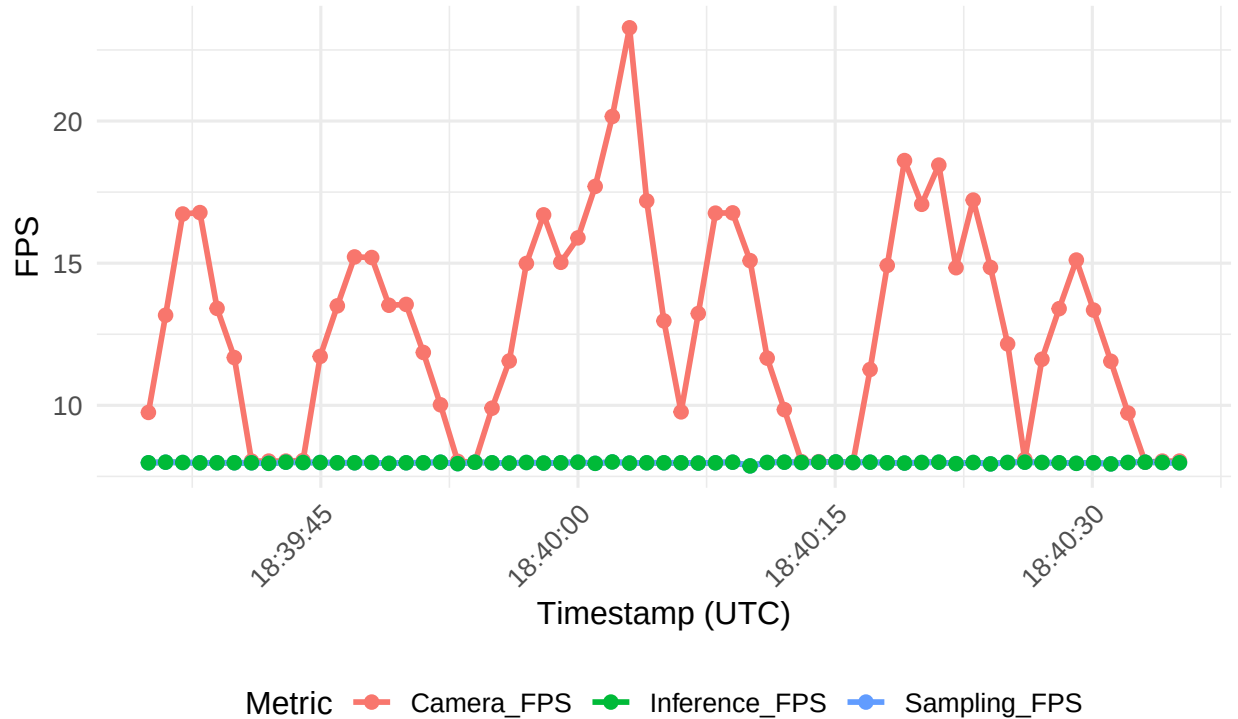
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Lubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Integrated Camera
- target sampling rate: 8
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline on laptop
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_sr8_20260922_213934.csv")
```

Performance Monitoring Metrics

File: fps_t420_sr8_20260922_213934.csv



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

Notes: **Camera_FPS** behaves in a weird & unexpected manner; Spikes reach over 20, pits are at inference/sampling fps. Rates. Inference latency range is huge (96 to 115), but a huge portion of it is around 100-105, with only a few exceptions at the tails. The results for this test show that it's better to keep using 7 fps sampling on laptop still. So that is my laptop baseline from now on. Next I will try whether or not changing from the integrated camera to an external webcam has any effect on anything. The hypothesis is no.

Test 2.4 T420, changing the camera

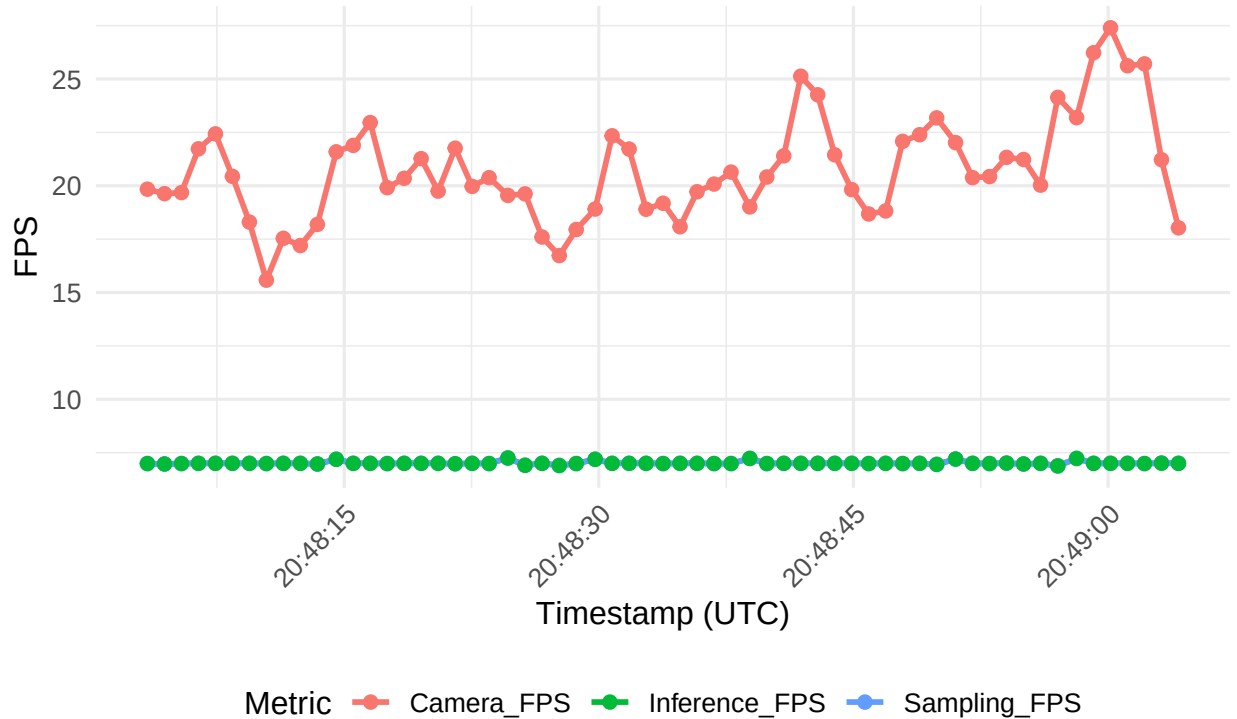
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 7
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: seeing the affect of changing the camera
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_logitech_base.csv")
```

Performance Monitoring Metrics

File: fps_t420_logitech_base.csv



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

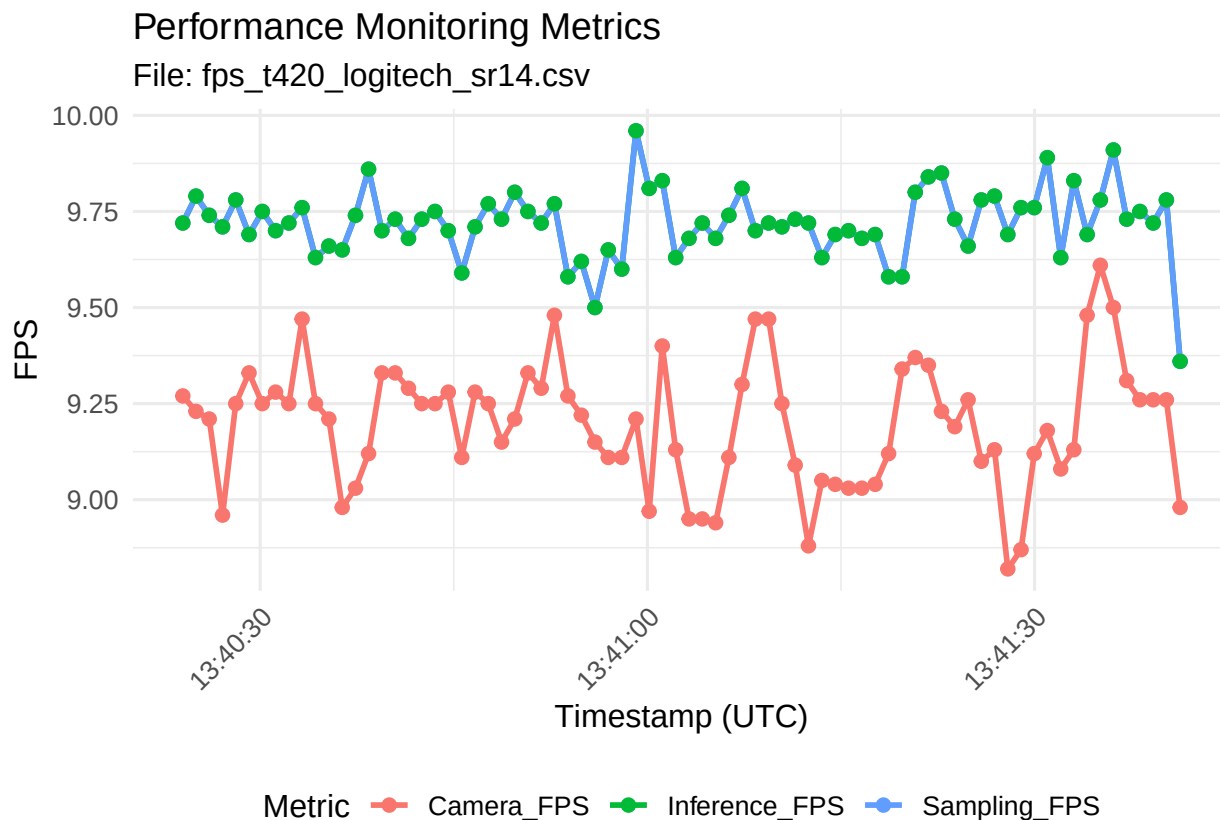
Notes: Contrary to my expectations, there is a wild jump in **Camera_FPS**. All values are now over 15 fps, meaning that sampling rate can be increased. **Latency_MS** ranges from 101 to 112. This prompts two new investigations: sr8 and sr14 on t420+logitech

Test 2.5 T420, trying higher framerate with camera

Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: attempting to establish new baseline on t420 with dedicated camera
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_logitech_sr14.csv")
```



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

Notes: As we expected, the hardware is unable to produce a sampling rate of 14, with **Sampling_FPS** moving around 9.75. But lowest tips are still above 9.25. Maybe it's worth to test if 9 is an acceptable sampling rate on t420+logitech, and only move test sr8 if we find 9 is not an acceptable baseline. For some reason **Camera_FPS** is below **Inference_FPS** now when we stress the system with a high sr target. The hypothesis that **Camera_FPS** is independent of the other systems seems to be wrong. **Latency_MS** ranges from 96 to 107, which would indicate an around 10 ms **Sampling_FPS**

Test 2.6 T420, attempting to establish sr9 baseline

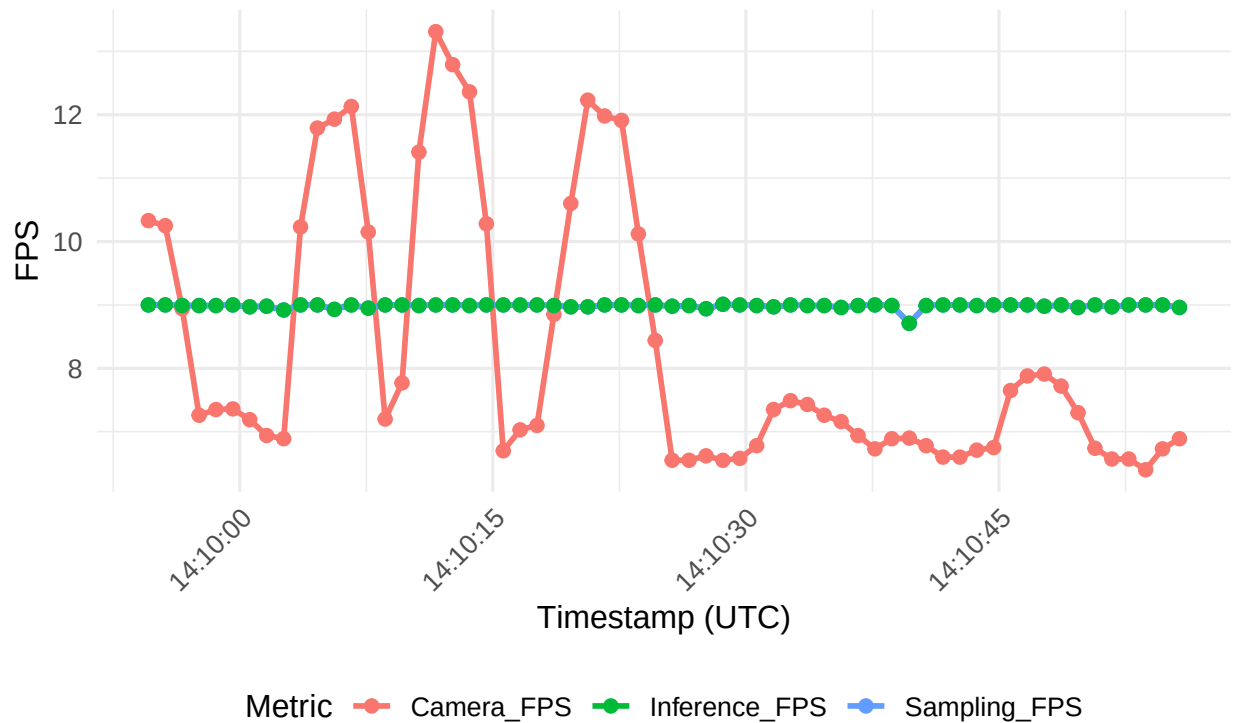
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 9
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: attempting to establish new baseline on t420 with dedicated camera
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_logitech_sr9.csv")
```

Performance Monitoring Metrics

File: fps_t420_logitech_sr9.csv



Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

Notes: Camera_FPS does weird fluctuation, occasionally going below Sampling_FPS. Thus sr9 is unacceptable. One more measurement on t420+logitech ought to be done with sr8.

Test 2.7 T420, attempting to establish sr8 baseline

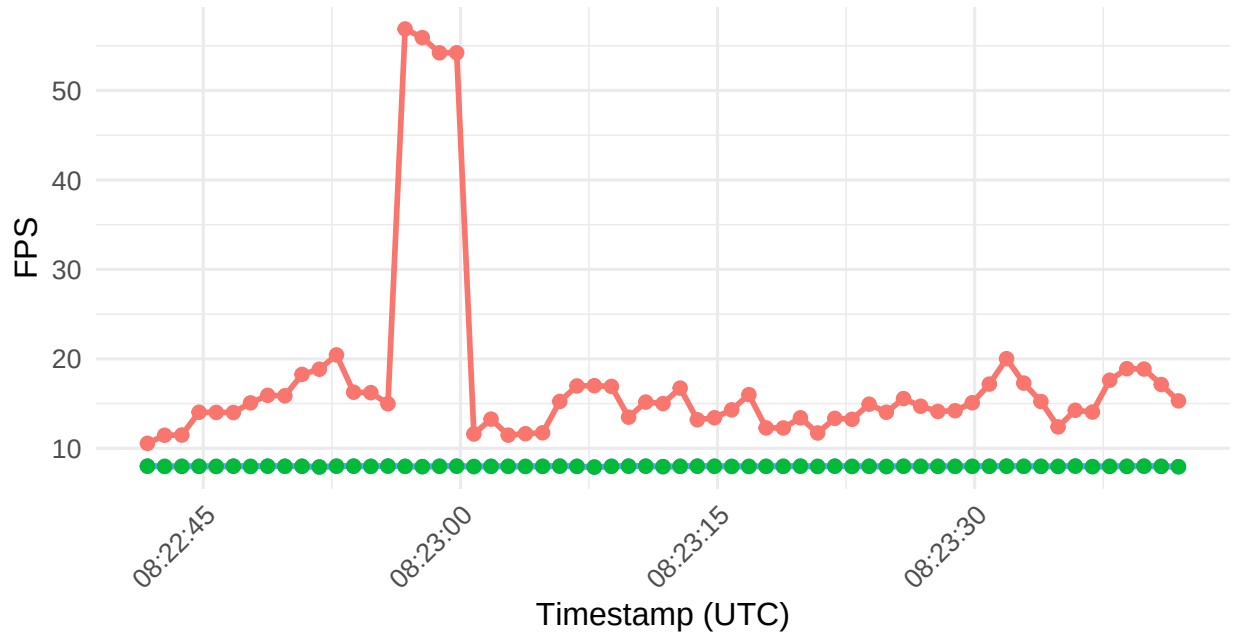
Environment

- device: Thinkpad T420 (Intel Core i5-2520M, 12GB RAM, no dedicated GPU)
- OS: Ubuntu 25.04
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:150.0) Gecko/20100101 Firefox/150.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 8
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: attempting to establish new baseline on t420 with dedicated camera
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_t420_logitech_sr8.csv")
```

Performance Monitoring Metrics

File: fps_t420_logitech_sr8.csv



Metric ● Camera_FPS ● Inference_FPS ● Sampling_FPS

Results

Nothing noteworthy in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

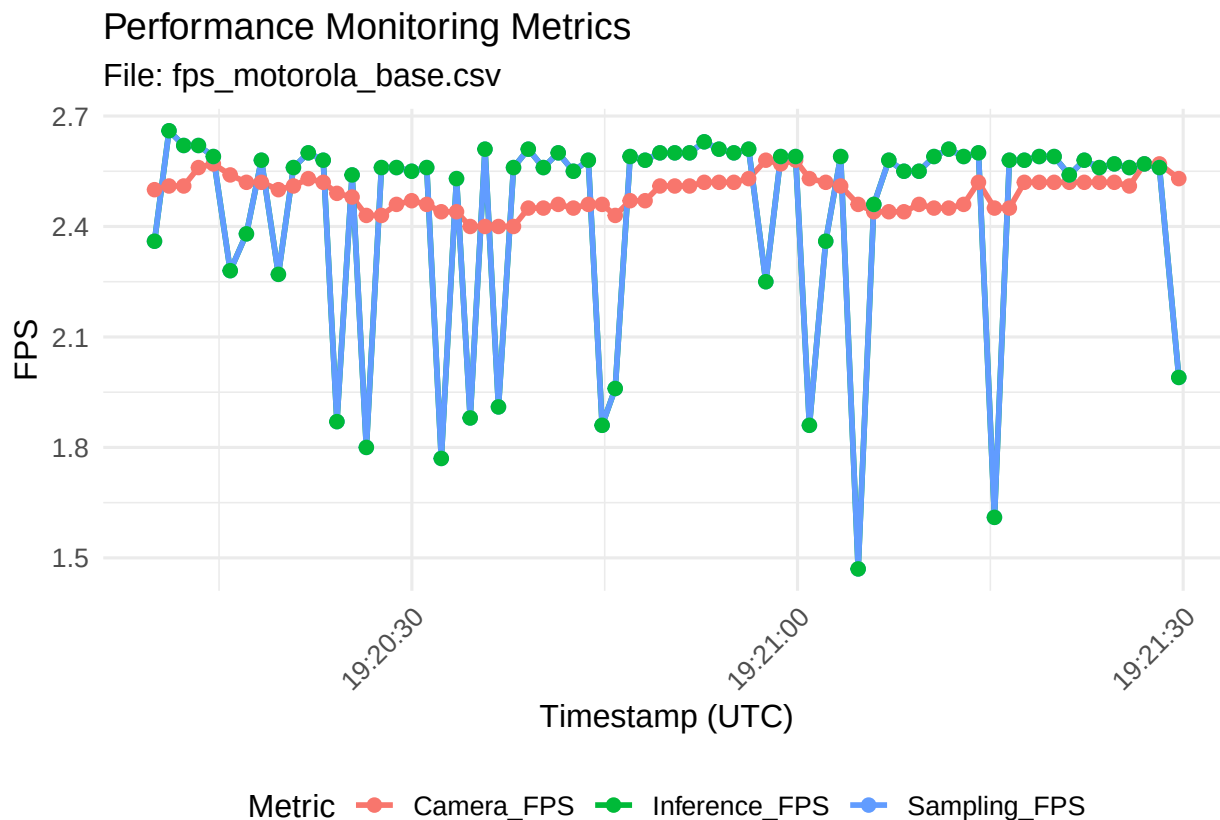
Notes: Camera_FPS continues to act weird, with a huge 50+ fps peak lasting a few seconds. Baselinewise: a sr8 keeps Inference_FPS and Sampling_fps at a constant 8, always below the camera curve. We now consider sr8 an acceptable baseline for t420 + logitech. Next up: measurements on a mobile device.

Test 2.8 Running on mobile

Environment

- device: Motorola moto g06 (MediaTek Heli G81 Extreme, 4GB+2GB(RAM boost), no dedicated GPU)
- OS: Android 15
- browser + version: Mozilla/5.0 (Android 15; Mobile; rv:157.0) Gecko/157.0 Firefox/157.0
- camera: Front 8M
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: Attempting to run application on mobile
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_motorola_base.csv")
```



Results

Can't access browser command line on mobile.

Notes: On mobile we're shockingly far away from our targets. **Camera_FPS** is between 2 and 3, while inference and sampling are between 1 and 3. I will decrease the target to 2, and then try to find a better baseline.

Test 2.9 Baseline on mobile

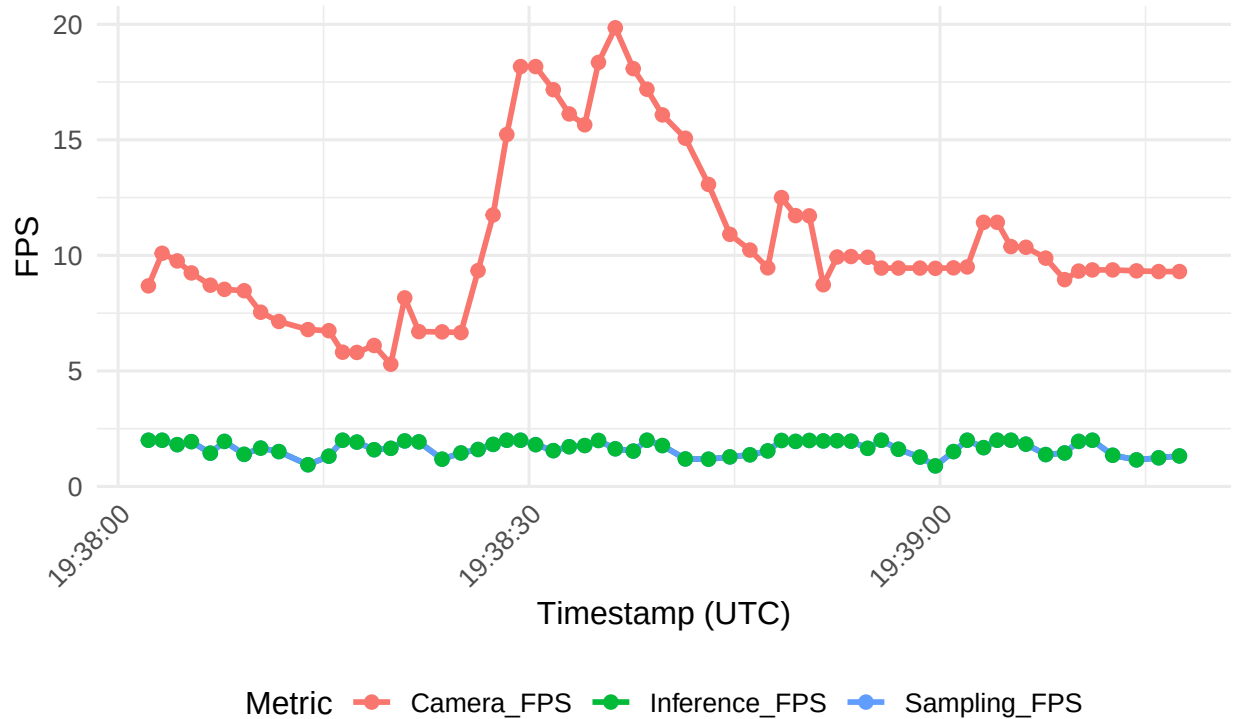
Environment

- device: Motorola moto g06 (MediaTek Heli G81 Extreme, 4GB+2GB(RAM boost), no dedicated GPU)
- OS: Android 15
- browser + version: Mozilla/5.0 (Android 15; Mobile; rv:157.0) Gecko/157.0 Firefox/157.0
- camera: Front 8M
- target sampling rate: 2
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: Attempting to find a baseline on mobile
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_motorola_sr2.csv")
```

Performance Monitoring Metrics

File: fps_motorola_sr2.csv



Results

Can't access browser command line on mobile.

Notes: Changing the SR target to 2 freed up the camera FPS. Now camera FPS moves between 5 and 20. Sampling & inference can't reach 2 in a stable fashion (multiple dips below). Nevertheless I think that it is still worth it to attempt SR3, possibly even higher if SR3 goes well.

Test 2.10 Baseline on mobile: SR3

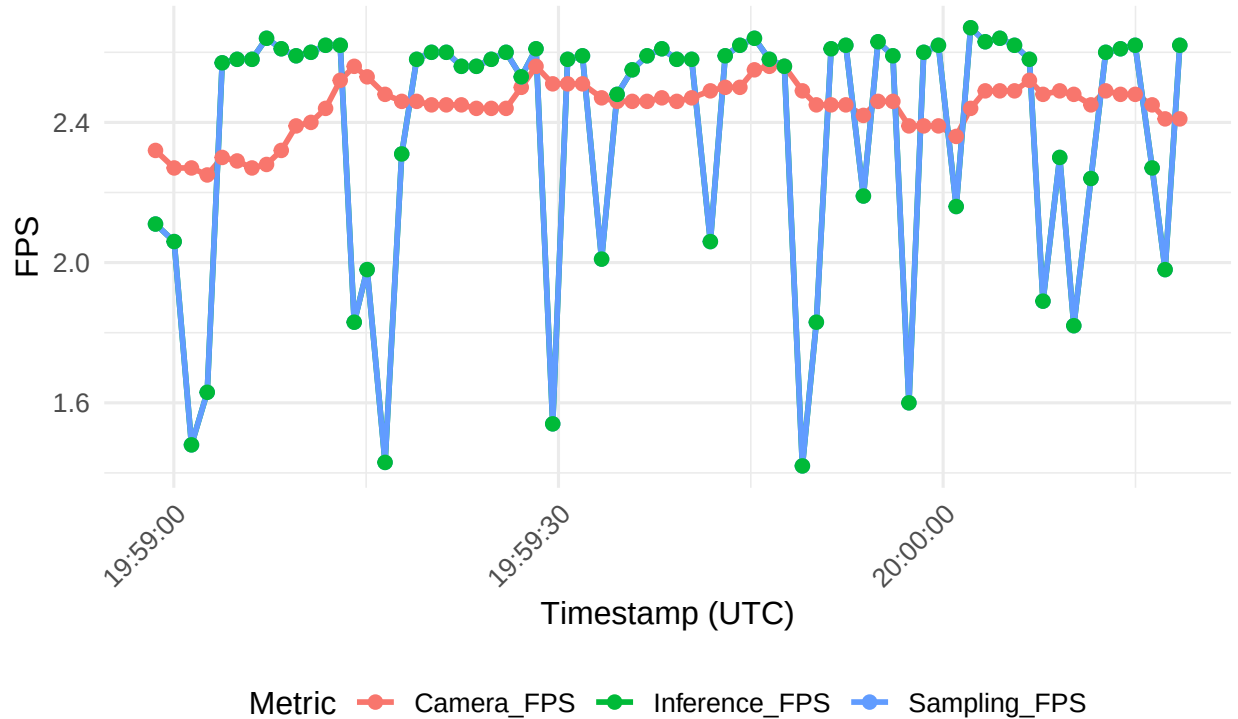
Environment

- device: Motorola moto g06 (MediaTek Heli G81 Extreme, 4GB+2GB(RAM boost), no dedicated GPU)
- OS: Android 15
- browser + version: Mozilla/5.0 (Android 15; Mobile; rv:157.0) Gecko/157.0 Firefox/157.0
- camera: Front 8M
- target sampling rate: 3
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: Attempting to find a baseline on mobile
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_motorola_sr3.csv")
```

Performance Monitoring Metrics

File: fps_motorola_sr3.csv



Results

Can't access browser command line on mobile.

Notes: We already hit the expected problems (trying to sample at a higher rate then what our latency allows for thus limiting the **Camera_FPS** and seldom reaching our sampling rate + causing wild fluctuations. Therefore we establish 2 as a mobile sample rate baseline.

Test series 3: Endurance

Test 3.1 stability over a long time

Environment

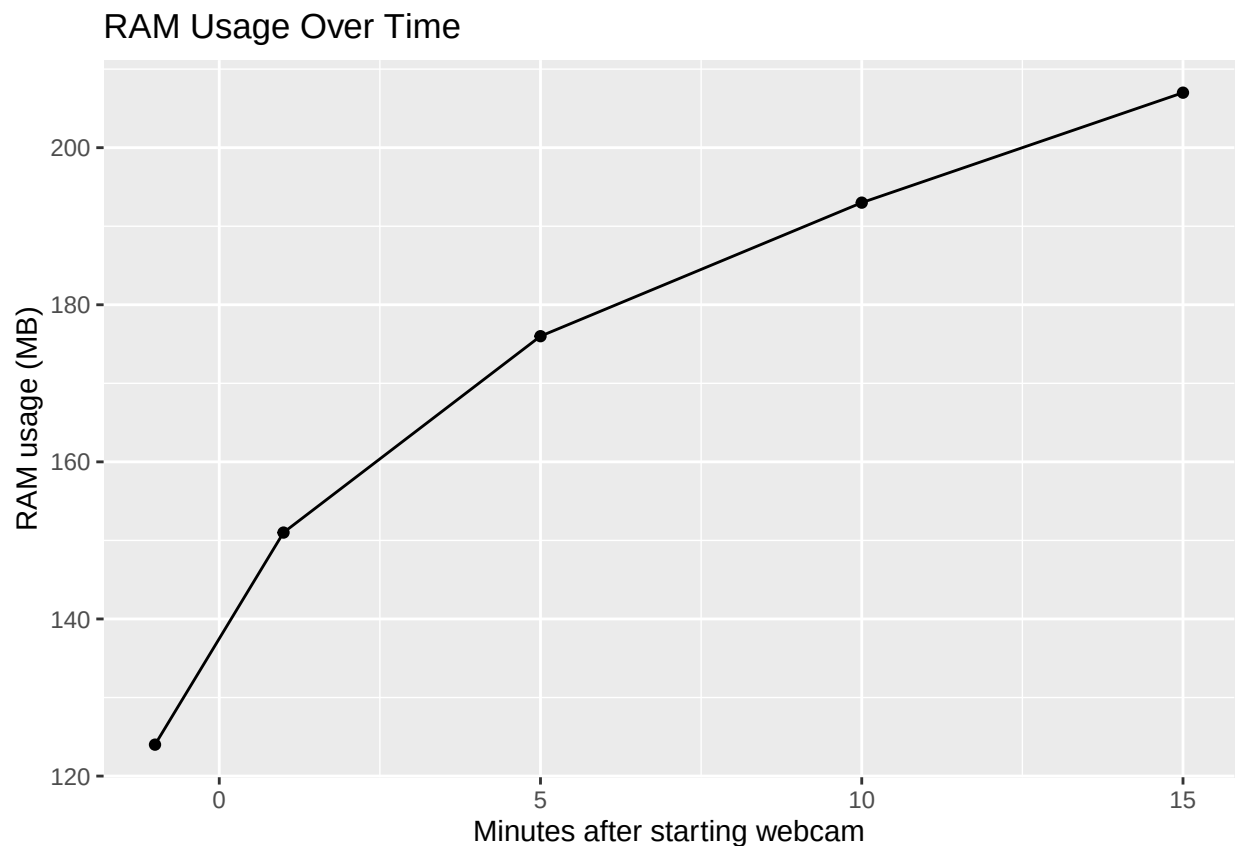
- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_l3f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: see & record any ram usage & misbehaviors over a long time
- duration of test: ~15 min
- special notes on env: -

Results I measured RAM usage of our app given around 15 minutes of webcam usage. **Camera_FPS**, Real sampling rate (vs target) etc. wasn't measured. Below are the results:

```
d <- data.frame(time=c(-1,1,5,10,15), value=c(124,151,176,193,207))
print(d)
```

```
##   time value
## 1   -1   124
## 2    1   151
## 3    5   176
## 4   10   193
## 5   15   207
```

```
ggplot(d, aes(time, value)) +
  geom_line() + geom_point() +
  labs(x="Minutes after starting webcam", y="RAM usage (MB)",
       title="RAM Usage Over Time")
```



Nothing noteworthy was observed in browser commandline. Only [Webcam inference] and [Webcam measurements] prints.

Browser responsiveness was also observed during this testing run with the browser remaining responsive. The only thing noteworthy is that keeping the tab open (Incl browser command line open + live camera feed etc) drastically raises cpu usage

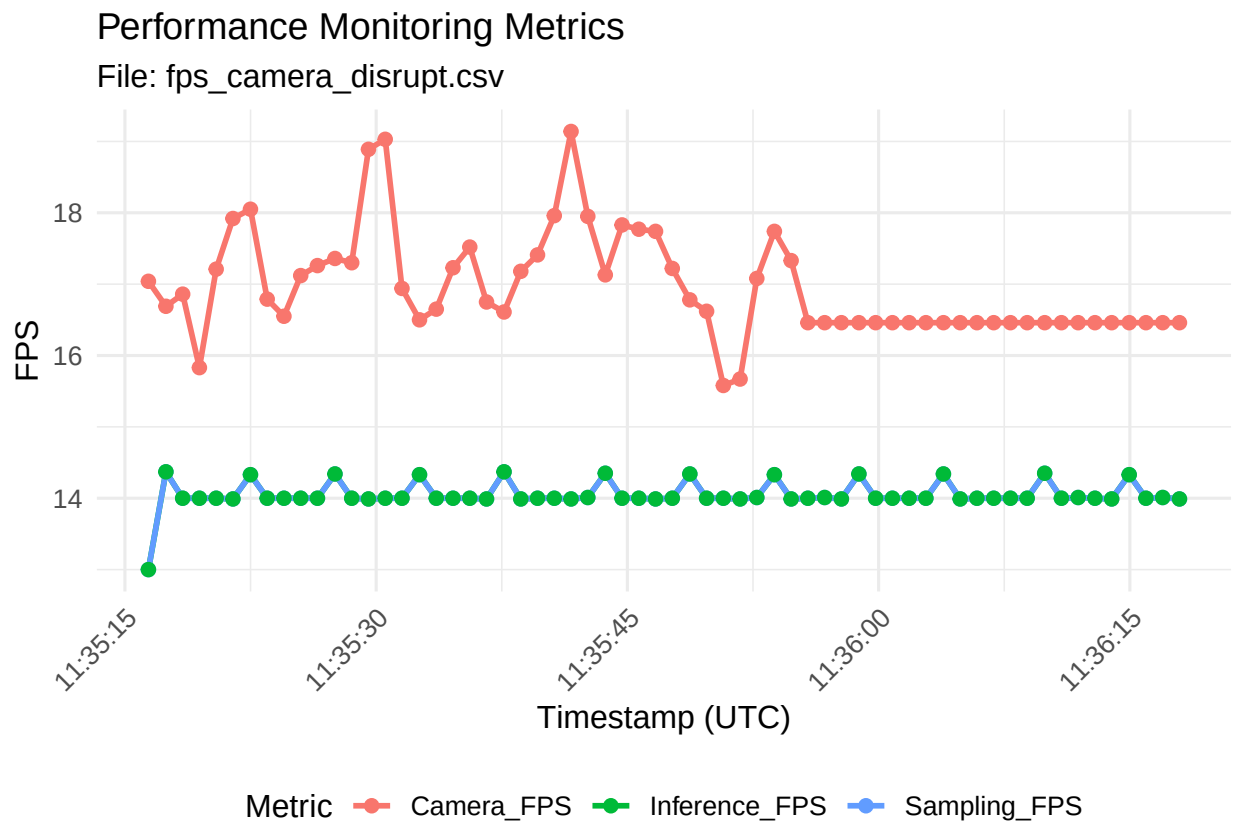
All in all even though we have constantly increasing memory usage & high CPU usage this around 200MB over 15min ram usage is acceptable for our use case even on less powerful devices.

Test 3.2 Reacting to camera disruptions

Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_l3f_20260807_225848`
- delegate: CPU
- score threshold: 0.35
- goal of test: see how the app reacts to the loss of the camera signal (achieved by physically unplugging the camera during a test)
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_camera_disrupt.csv")
```



Results

No new frames incoming: last known camera frame frozen on `<video id="camera-video">`.

NOTABLY: nothing on interface (like generic status print, camera status print, webcam feed) OR the browser command line indicates camera disruption. Underlying systems continue with no more frames coming in from sampling.

Moment of camera disruption clearly visible on graph: **Camera_FPS** reporting doesn't update the number after no new frames available. It permanently stays *16.46FPS*.

Plugging the camera back in while the test is still ongoing doesn't autoresume frame sampling etc. Possible weakness in app found.

Test 3.3 Reacting to start-stop-loops

Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: see how the app reacts to repeatedly turning the camera on and off
- duration of test: ~1 min
- special notes on env: -

Results No notable effect on inference rates, sampling rates, `Camera_FPS`. No unusual prints on browser command line or status renderings. Measurement recording malfunctions due to camera stop also stopping it. This is intended behavior. No new weaknesses to report here.